# CSGE602055 Operating Systems
# CSF2600505 Sistem Operasi
# Week 03: File System & FUSE

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https://os.vlsm.org/Slides/os03.pdf
Always check for the latest revision!

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# OS232[3]): Operating Systems Schedule 2023 - 2

| Week | Topic[1]) | OSC10[2]) |
|---|---|---|
| Week 00 | Overview (1), Assignment of Week 00 | Ch. 1, 2 |
| Week 01 | Overview (2), Virtualization & Scripting | Ch. 1, 2, 18. |
| Week 02 | Security, Protection, Privacy, & C-language. | Ch. 16, 17. |
| Week 03 | File System & FUSE | Ch. 13, 14, 15. |
| Week 04 | Addressing, Shared Lib, & Pointer | Ch. 9. |
| Week 05 | Virtual Memory | Ch. 10. |
| Week 06 | Concurrency: Processes & Threads | Ch. 3, 4. |
| Week 07 | Synchronization & Deadlock | Ch. 6, 7, 8. |
| Week 08 | Scheduling + W06/W07 | Ch. 5. |
| Week 09 | Storage, Firmware, Bootloader, & Systemd | Ch. 11. |
| Week 10 | I/O & Programming | Ch. 12. |

[1]) For schedule, see https://os.vlsm.org/#idx02

[2]) Silberschatz et. al.: **Operating System Concepts**, $10^{th}$ Edition, 2018.

[3]) This information will be on **EVERY** page two (2) of this course material.

# STARTING POINT — https://os.vlsm.org/

- **Text Book** — Any recent/decent OS book. Eg. (**OSC10**) Silberschatz et. al.: **Operating System Concepts**, $10^{th}$ Edition, 2018. (See https://www.os-book.com/OS10/).
- **Resources** (https://os.vlsm.org/#idx03)
  - **SCELE OS232** — https://scele.cs.ui.ac.id/course/view.php?id=3614. The enrollment key is **XXX**.
  - **Download Slides and Demos from GitHub.com** — (https://github.com/os2xx/os/) os00.pdf (W00), os01.pdf (W01), os02.pdf (W02), os03.pdf (W03), os04.pdf (W04), os05.pdf (W05), os06.pdf (W06), os07.pdf (W07), os08.pdf (W08), os09.pdf (W09), os10.pdf (W10).
  - **Problems** 195.pdf (W00), 196.pdf (W01), 197.pdf (W02), 198.pdf (W03), 199.pdf (W04), 200.pdf (W05), 201.pdf (W06), 202.pdf (W07), 203.pdf (W08), 204.pdf (W09), 205.pdf (W10).
  - **LFS** — http://www.linuxfromscratch.org/lfs/view/stable/
  - **OSP4DISS** — https://osp4diss.vlsm.org/
  - **This is How Me Do It!** — https://doit.vlsm.org/
    - PS: "Me" rhymes better than "I", duh!

# Agenda


1. Start
2. OS232 Schedule
3. Agenda
4. Week 03
5. OSC10 (Silberschatz) Chapter 13, 14, and 15
6. File System Interface
7. File System Organization
8. FHS: Filesystem Hierarchy Standard
9. Devices
10. File System Implementation
11. File System Internals

# Week 03 File System & FUSE: Topics[1]

- Files: data, metadata, operations, organization, buffering, sequential, nonsequential
- Directories: contents and structure
- File systems: partitioning, mount/unmount, virtual file systems
- Standard implementation techniques
- Memory-mapped files
- Special-purpose file systems
- Naming, searching, access, backups
- Journaling and log-structured file systems

[1] Source: ACM IEEE CS Curricula

# Week 03 File System & FUSE: Learning Outcomes[1]

- Describe the choices to be made in designing file systems. [Familiarity]
- Compare and contrast different approaches to file organization, recognizing the strengths and weaknesses of each. [Usage]
- Summarize how hardware developments have led to changes in the priorities for the design and the management of file systems. [Familiarity]
- Summarize the use of journaling and how log-structured file systems enhance fault tolerance. [Familiarity]

[1] Source: ACM IEEE CS Curricula

# OSC10 (Silberschatz) Chapter 13: File-System Interface, Chapter 14: File System Implementation, and Chapter 15: File System Internals

- OSC10 Chapter 13
  - File Concept
  - Access Methods
  - Disk and Directory Structure
  - Protection
  - Memory-Mapped Files
- OSC10 Chapter 14
  - File-System Structure
  - File-System Operations
  - Directory Implementation
  - Allocation Methods
  - Free-Space Management
  - Efficiency and Performance
  - Recovery
  - Example: WAFL File System
- OSC10 Chapter 15
  - File Systems
  - File-System Mounting
  - Partitions and Mounting
  - File Sharing
  - Virtual File Systems
  - Remote File Systems
  - Consistency Semantics
  - NFS

# File System Interface

- File Concept
  - File Attributes: Name, Id, Type, Location, Size, Protection, Time Stamp: create, last modified, last accessed.
  - File Operation
    - Create/Delete/Truncate
    - Open/Close
    - Read/Write
  - File Types: Executable, Object, Source Code, Library, Markup, Markdown, Archive, Compressed.
  - File Structure: No Structure (just a string).
  - Access Methods: Sequential vs Direct Access
- Directory and Disk Structure
  - Three-Structured Directories
  - Directory Operation: create/delete, search/list, rename, traverse
  - Path Name: Absolute vs. Relative
  - FS Mounting vs. Volume Based System
- File Sharing
- Protection: Access Control (eg. -rwx–x–x)

# File System Organization

- Disk Partition
  - One Disk — Many Partitions
  - Many Disks — One Partitions
  - Many Disks — Many Partitions
  - One Partition — One File System (Volume)
- Mounting vs. Volumes

```
demo@badak:~$ df
Filesystem      1K-blocks      Used Available Use% Mounted on
/dev/sda2         9515660   1435776   7573468  16% /
/dev/sdb1        32895760  12156672  19045036  39% /usr
/dev/sdc1       412322216  79695252 311639116  21% /home
udev                10240         0     10240   0% /dev
tmpfs            16508828         0  16508828   0% /dev/shm
tmpfs             6603532      8880   6594652   1% /run
tmpfs                5120         0      5120   0% /run/lock
tmpfs            16508828         0  16508828   0% /sys/fs/cgroup
tmpfs             3301768         0   3301768   0% /run/user/1002
demo@badak:~$
```

# FHS: Filesystem Hierarchy Standard

- Source (URL) http://refspecs.linuxfoundation.org/FHS_3.0/fhs-3.0.pdf
- A file placement guidelines/requirements for GNU/Linux-like OS.

| FILES | shareable (multiple hosts) | unshareable (single hosts) |
|---|---|---|
| static (read only, except for update) | /usr, /opt | /etc, /boot |
| variable (r/w) | /var/mail, /var/spool/news | /var/run, /var/lock |

- The Root File System (Required)

| Directory | Description |
|---|---|
| /bin | Essential command binaries |
| /boot | Static files of the boot loader |
| /dev | Device files |
| /etc | Host-specific system configuration |
| /lib | Essential shared libraries and kernel modules |
| /media | Mount point for removable media |
| /mnt | Mount point for mounting a filesystem temporarily |
| /opt | Add-on application software packages |
| /run | Data relevant to running processes |
| /sbin | Essential system binaries |
| /srv | Data for services provided by this system |
| /tmp | Temporary files |
| /usr | Secondary hierarchy |
| /var | Variable data |

# More FHS 1

- Specific Options

| Directory | Description |
|---|---|
| /home | User home directories (optional) |
| /lib<qual> | Alternate format essential shared libraries(optional) |
| /root | Home directory for the root user (optional) |

- The /usr Hierarchy

| Directory | Description |
|---|---|
| /usr/bin | Most user commands (required) |
| /usr/lib | Libraries (required) |
| /usr/local | Local hierarchy (empty after main installation) (required) |
| | /usr/local/{bin\|etc\|games\|include\|lib\|man\|sbin\|share\|src} (required) |
| /usr/sbin | Non-vital system binaries (required) |
| /usr/share | Architecture-independent data (required) |
| | /usr/share/{man\|misc} (required) |
| | /usr/share/{color\|dict\|doc\|games\|info\|locale} (optional) |
| | /usr/share/{nls\|ppd\|sgml\|terminfo\|tmac\|xml\|zoneinfo} (optional) |
| /usr/games | Games and educational binaries (optional) |
| /usr/include | Header files included by C programs (optional) |
| /usr/libexec | Binaries run by other programs (optional) |
| /usr/lib<qual> | Alternate Format Libraries (optional) |
| /usr/src | Source code (optional) |

# More FHS 2

- The /var Hierarchy

| Directory | Description |
|---|---|
| /var/cache | Application cache data (required) |
| /var/lib | Variable state information (required) |
| | /var/lib/misc (required) |
| /var/local | Variable data for /usr/local (required) |
| /var/lock | Lock fileslogLog files and directories (required) |
| /var/opt | Variable data for /opt (required) |
| /var/run | Data relevant to running processes (required) |
| /var/spool | Application spool data (required) |
| /var/tmp | Temporary files preserved between system reboots (required) |
| /var/backups | (reserved names, do not use) |
| /var/cron | (reserved names, do not use) |
| /var/msgs | (reserved names, do not use) |
| /var/preserve | (reserved names, do not use) |
| /var/account | Process accounting logs (optional) |
| /var/crash | System crash dumps (optional) |
| /var/games | Variable game data (optional) |
| /var/mail | User mailbox files (optional) |
| /var/yp | Network Information Service (NIS) database files(optional) |

# More FHS 3

- (Mostly) Linux

| Directory | Description |
|---|---|
| /proc | Kernel and process information virtual filesystem |
| /sys | Kernel and system information virtual filesystem |
| /usr/include | Header files included by C programs |
| /usr/src | Source code |
| /var/spool/cron | cron and at jobs |

# Devices

- the `/dev/` directory
  - `/etc/fstab`: configuration of filesystems
  - `/etc/mtab` → `/proc/mounts`: mounted filesystems
  - `/proc/swaps`: swap filesystems
  - `df`: checking diskspace and filesystems
  - Device Major and Minor Numbers
  - UUID - Universally Unique IDentifier (128 bits)
  - GUID - Globally Unique IDentifiers: `ls -al /dev/disk/by-uuid`
  - practically is NOT guaranteed unique
  - FUSE: Filesystem in Userspace
  - More Storage Structure
    - `tmpfs` — a temporary file storage, stored in RAM that grows and shrinks.
    - `objfs` — dynamic kernel object filesystem.
    - `ctfs` — (creating, controlling, and observing) contract file system .
    - `loopfs` — loop filesystem allows to dynamically allocate loop devices.
    - `procfs` — proc filesystem presents information about processes.
    - `ufs` — the original Unix Filesystem (before Linux ext2).
    - `zfs` — the Zettabyte Filesystem is both a volume manager and a file system.

# A Typical Ubuntu 20.04 Work Station

```
cbkadal@ubuntu2004:~$ df
Filesystem      1K-blocks       Used Available Use% Mounted on
udev              8138664          0   8138664   0% /dev
tmpfs             1634140       1948   1632192   1% /run
tmpfs             8170684     210348   7960336   3% /dev/shm
tmpfs                5120          4      5116   1% /run/lock
tmpfs             8170684          0   8170684   0% /sys/fs/cgroup
tmpfs             1634136         76   1634060   1% /run/user/1000
/dev/sda1           98304      33523     64781  35% /boot/efi
/dev/sda3       286082372   78565916 207516456  28% /altfs/ntfs
/dev/sda5        32999120    9181772  22111364  30% /altfs/linux1
/dev/sda6        38186548   12054612  24162428  34% /altfs/linux2
/dev/sda7       126265680   13342928 106465768  12% /
/dev/sdb2        62216964   13238156  45788588  23% /var
/dev/sdb3      3532259904 2605226568 747535200  78% /home
/dev/loop0         101632     101632         0 100% /snap/core/10859
/dev/loop1          65920      65920         0 100% /snap/gtk-common-themes/1513
/dev/loop2          66432      66432         0 100% /snap/gtk-common-themes/1514
/dev/loop3         678016     678016         0 100% /snap/intellij-idea-community/273
/dev/loop4         679040     679040         0 100% /snap/intellij-idea-community/270
/dev/loop5          52352      52352         0 100% /snap/snap-store/498
/dev/loop6         223232     223232         0 100% /snap/gnome-3-34-1804/60
/dev/loop7         267008     267008         0 100% /snap/kde-frameworks-5-core18/32
/dev/loop8         166784     166784         0 100% /snap/gnome-3-28-1804/145
/dev/loop9         102784     102784         0 100% /snap/kotlin/57
/dev/loop10         52352      52352         0 100% /snap/snap-store/518
/dev/loop11         56832      56832         0 100% /snap/core18/1988
###########        ##### TL;DR #####         # #### #################
/dev/loop18         56832      56832         0 100% /snap/core18/1944
/dev/loop19        142080     142080         0 100% /snap/chromium/1506
```

# File Systems Implementation

- File System Layers / Structure
  - Application Programs
  - Logical File Systems
  - File-Organization Module
  - Basic File Systems
  - I/O Control
  - Hardware Device
- File System Implementation
- File Control Block
- FS In Memory Structure
- VFS: Virtual File Systems
  - How to support multiple File Systems
  - I.e. How to support multiple `open()/close()` `read()/write()` operations

# Implementation and Allocation Method

- Directory Implementation
  - Linear List
  - Hash Table
- Allocation Method
  - Contiguous
  - Linked
  - Indexed
  - Combined Scheme
- Free Space Management
- Performance & Efficiency
- Unified Buffer Cache
- Recovery
- Log Structured File System

# File Systems Internals

- File Systems
- File-System Mounting
- Partitions and Mounting
- File Sharing
- Virtual File Systems
- Remote File Systems
- Consistency Semantics
- NFS